

Traffic Signal Violation Detection at Simpang Gedongan Sleman Using YOLOv8, HSV-Based Segmentation, and DEEPSORT

1st Deira Aisya Refani

24/532821/PA/22539

Universitas Gadjah Mada

Department of Computer Science and Electronics

Yogyakarta, Indonesia

deiraaisyarefani532821@mail.ugm.ac.id

2nd Finanazwa Ayesha

24/532953/PA/22556

Universitas Gadjah Mada

Department of Computer Science and Electronics

Yogyakarta, Indonesia

finanazwaayesha@mail.ugm.ac.id

3rd Farrel Tsaqif Anindyo

24/536735/PA/22769

Universitas Gadjah Mada

Department of Computer Science and Electronics

Yogyakarta, Indonesia

farreltsaqifanindyo@mail.ugm.ac.id

4th Yohana Butar Butar

24/546690/PA/23212

Universitas Gadjah Mada

Department of Computer Science and Electronics

Yogyakarta, Indonesia

yohanabutarbutar546690@mail.ugm.ac.id

Abstract—This is a placeholder abstract. Replace this text with a concise summary of the research background, method, main findings, and conclusion.

I. INTRODUCTION

A. Background

Write the background of the problem here.

B. Problem Statement

Write the problem statement here.

C. Objective

Write the main objective of the study here.

D. Scope

Write the scope and limitations of the study here.

II. LITERATURE REVIEW

A. Related Work

Write the related work summary here.

B. Key Concepts

Write the key concepts, theories, or methods here.

III. METHODOLOGY

A. Proposed Approach

Write the overall system design or workflow here.

B. Dataset

Write the dataset description here.

C. Preprocessing

Write the preprocessing steps here.

D. Detection and Tracking

Write the detection and tracking method here.

E. Violation Logic

Write the violation decision logic here.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

Write the experimental setup here.

B. Results

Write the main results here.

C. Discussion

Write the interpretation of the results here.

V. CONCLUSION

A. Conclusion

Write the final conclusion here.

B. Future Work

Write possible future improvements here.

REFERENCES